ABCPROOF · MOTIZUKIPDF · SPECIAL BROADCAST

# funfun IUT

An otaku field guide to inter-universal Teichmüller theory （宇宙際タイヒミューラー理論）: its characteristics, some honest observations, and far too many Japanese pop culture jokes

Running example: the abc triple $(1, 2400, 2401)$ with $l = 13$ · all numbers from the toolkit in `src/` · November 2026

**ABSTRACT** Inter-universal Teichmüller theory (IUT) is Shinichi Mochizuki's 73-process, four-paper, 646-page machine for bounding the height of an elliptic curve by its conductor, which is to say for proving the abc conjecture. We describe what kind of theory it is, using the papers in `pdf/` and the data-type toolkit in this repository, and we follow one small example through it. Along the way we compare its objects to things from anime, manga, games and Japanese daily life, because a theory with "theaters", "pilots", "alien copies" and "Hodge theaters in different universes" is already halfway to being a light novel. No mathematical claim below rests on a joke, and no joke rests on a mathematical claim.

**CONTENT NOTICE**

All references are text only: no character art, logos or screenshots were harmed, or used, in the making of this article. Where we describe what the theory *claims*, we cite the papers. Where we describe the debate about whether the claim is *right*, we report both sides and adjudicate nothing. We are a toolkit, not a referee. 審判ではありません。

## 0 Previously on IUT 前回までのあらすじ

In August 2012 four preprints appeared on a RIMS (Kyoto University) web page with no press release: *Inter-universal Teichmüller Theory I–IV* [IUTchI–IV], 186 + 174 + 199 + 87 pages. They were accepted by the journal PRIMS on February 5, 2020, announced at a Kyoto University press conference on April 3, 2020, and published in special volumes of PRIMS in 2021 [EssLgc, §1.1]. In March 2018, Peter Scholze and Jakob Stix spent a week at RIMS discussing the theory with Mochizuki and came away convinced that the key inequality, [IUTchIII] Corollary 3.12, does not follow; Mochizuki came away convinced that their objection rests on a misreading he calls the identification of "redundant copies" [EssLgc, §1]. Since then there have been surveys [Alien], a logic-focused exposition [EssLgc], effective versions [ExpEst], and in April 2026 a progress report on formalizing the theory in Lean [Form26].

If this sounds like a long-running shōnen series, it is. The first arc aired in 2012, the tournament arc started in 2018, a movie adaptation (the published version) came out in 2021, and the fandom is still arguing about whether the finale was earned. Unlike most long-running series, the main character has not yet had a power-up beach episode, although [Alien] comes close: it explains the whole theory through the Gaussian integral $\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$.

## 1 The plot in one page あらすじ

The abc conjecture is about coprime positive integers with $a + b = c$. Write $\mathrm{rad}(abc)$ for the product of the distinct primes dividing $abc$, and define the *quality* of the triple as

$$q(a,b,c) \;=\; \frac{\log c}{\log \mathrm{rad}(abc)}.$$

The conjecture says that for every $\varepsilon > 0$ only finitely many triples have $q > 1 + \varepsilon$. Our running example is

$$1 + 2400 = 2401, \qquad 2400 = 2^5 \cdot 3 \cdot 5^2, \quad 2401 = 7^4, \qquad \mathrm{rad}(abc) = 2 \cdot 3 \cdot 5 \cdot 7 = 210,$$

so $q = 7.7836/5.3471 = 1.4557$. That is a respectable triple. (The current champion, $2 + 3^{10} \cdot 109 = 23^5$, has quality about 1.6299.) Its $c$ is also what you get if you collect all seven dragon balls four times over: $7 \cdot 7 \cdot 7 \cdot 7$. We wished for a small radical and got one.

IUT does not attack the integers directly. It turns the triple into the Frey curve $E : y^2 = x(x-1)(x+2400)$ and proves a Szpiro / Vojta type inequality for elliptic curves over number fields: in the form of [IUTchIV] Corollary 2.3, for a hyperbolic curve $U_X = X \setminus D$,

$$\mathrm{ht}_{\omega_X(D)} \;\lesssim\; (1+\varepsilon)\,\big(\text{log-diff}_X + \text{log-cond}_D\big),$$

where $\lesssim$ means "up to a function bounded below by a constant", i.e. an inequality of *bounded discrepancy classes*. Taking $X = \mathbb{P}^1$ and $D = \{0, 1, \infty\}$ gives abc. In the toolkit's normalization, the example has

$$\log(\underline{q}) = 24.2023, \qquad |\log(\underline{q})| = \tfrac{\log(q)}{2l} = 0.9309, \qquad \mathrm{ht}_E = \tfrac{1}{6}\log(\underline{q}) = 4.0337, \qquad \log(\mathfrak{f}) = 5.3471.$$

Here $\log(\underline{q})$ is the arithmetic degree of the $q$-parameters at the bad places $v_2, v_3, v_5, v_7$, where $\mathrm{ord}(q) = 2, 2, 4, 8$, and $\log(\mathfrak{f}) = \log 210$ is the conductor. The prime $l = 13$ is the auxiliary prime of the initial $\Theta$-data [IUTchI, Def 3.1]; it sets the size of everything that follows.

> **OTAKU NOTE: THE PROTAGONIST'S STATS**
> Every isekai protagonist checks their status window on arrival. Ours reads: **Class** Frey curve. **Level** $l = 13$. **Labels** $\mathbb{F}_l = \{-6, \ldots, 6\}$, with $l^* = (l-1)/2 = 6$ and $l^{\pm} = (l+1)/2 = 7$. **Party** 6 valuations $v_2, v_3, v_5, v_7, v_{13}, v_\infty$. **Equipment** a field $F$ with $[F : \mathbb{Q}] \le 276480$ and $K = F(E[l])$ with $[K : \mathbb{Q}] \le 7245987840$. **Special skill** none yet; it has to be built, one Hodge theater at a time.

## 2 Characteristic one: it is an isekai 異世界もの

The word "inter-universal" is literal. Each of the main objects, a *$\Theta^{\pm ell}$NF-Hodge theater*, is built inside its own Grothendieck universe, and the central construction, the *$\Theta$-link*, connects Hodge theaters that live in *different* universes. The $\Theta$-link is not a ring homomorphism. It does not even pretend to be one: it identifies only certain multiplicative monoids and the realified global Frobenioids attached to them, sending the theta values on one side to the $q$-parameter on the other,

$$^{\dagger}\mathcal{F}_{\mathrm{LGP}}^{\mathbb{H}\blacktriangleright\times\mu} \;\xrightarrow{\sim}\; {}^{\ddagger}\mathcal{F}_{\triangle}^{\mathbb{H}\blacktriangleright\times\mu}, \qquad \underline{\underline{\Theta}}_v^{\,j^2} \;\longmapsto\; \underline{\underline{q}}_v \quad (1 \le j \le l^*)$$

[IUTchIII, Def 3.8; IUTchII, Cor 4.10]. The additive structure does not cross. In other words, when you are hit by Truck-kun and reincarnated on the other side of a Θ-link, you keep your multiplicative monoid, your units modulo torsion and your Frobenioid-theoretic degree, and you lose the ability to add. This is the "Reincarnated as a Monoid" route, and it is the whole point: $\underline{\underline{q}}_v^{j^2}$ and $\underline{\underline{q}}_v$ can be identified across universes precisely because nobody on either side insists that they are the same *ring* element.

Because the two sides are not related by any ring isomorphism, the objects on one side are, from the point of view of the other, *alien copies* [Alien, §1.3, §2.10]. The theory then has to answer the question every isekai story eventually asks: what can the new world know about the old one? Its answer is "only what can be reconstructed from data that both worlds share". The shared data are called *coric*, and they are the subject of §5.

> **OTAKU NOTE: THE GUILD REGISTRATION**
> In a fantasy guild you register your stats, not your soul. A Θ-link is the same: the guild clerk on the ‡ side writes down "multiplicative monoid, units mod torsion, degree" and stamps the form. Your additive structure is not on the form. Please do not argue with the clerk about it. Many have tried.

## 3 Characteristic two: it takes rings apart 分解と合体

A ring has two operations, and in a number field they are tangled together so tightly that you cannot change one without the other. Mochizuki describes a ring as having *two underlying combinatorial dimensions*, roughly its additive and multiplicative structures, and calls the tangle the *arithmetic holomorphic structure* [Alien, §2.7, §2.11]. The strategy of IUT is to *dismantle* this intertwining [Alien, §2.7, §2.10], move the pieces around separately, and then measure how much was lost.

Two moves do the moving. The Θ-link (horizontal) keeps only multiplicative data, as above. The *log-link* (vertical) applies the $p$-adic logarithm at each place $v$,

$$\log_v : \mathcal{O}_{K_v}^{\times} \longrightarrow \overline{K}_v, \qquad \mathcal{I}_v = (p_v^*)^{-1} \cdot \log_v\big(\mathcal{O}_{K_v}^{\times}\big),$$

turning multiplication into addition. The image $\mathcal{I}_v$ is the *log-shell*, where $p_v^* = p_v$ for odd $p_v$ and $p_v^* = 4$ for $p_v = 2$ (the toolkit prints $p^* = 4, 3, 5, 7, 13$ for the example). [Alien] describes the log-link as an "arithmetic rotation, or juggling" of the two dimensions, and [EssLgc] says the log-link is "a device for constructing additive log-shells".

This is, unmistakably, a combining-robot show. In episode one the giant robot separates into a jet, a tank and a submarine; each is useful on its own but none can win the fight; the climax is the transformation sequence where they recombine. IUT separates the ring into its additive part (the jet, fast and flexible, lives in log-shells) and its multiplicative part (the tank, heavy, crosses Θ-links), and then spends [IUTchIII] working out how to recombine them in a universe where the original robot was never built. The difference from the anime is that the recombination is only *up to indeterminacy*, which is like the robot coming back with its arms a little bit fuzzy.

## 4 Characteristic three: the board is a lattice すごろく盤

Iterating both moves gives the *log-theta-lattice* [IUTchIII, Def 1.4, 3.8]: a Hodge theater ${}^{n,m}\mathcal{HT}$ at every point $(n, m) \in \mathbb{Z}^2$, with a Θ-link $(n, m) \to (n+1, m)$ and a log-link $(n, m) \to (n, m+1)$. The toolkit draws 10

vertices and 9 links for two columns and five rows:

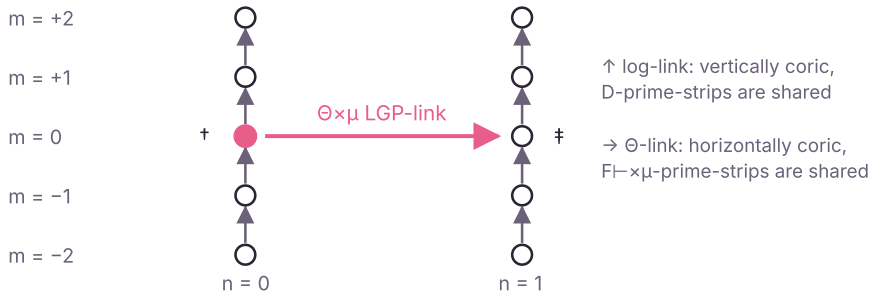



**Figure 1.** Part of the log-theta-lattice as the toolkit builds it (`iut walk --rows 2`). Each dot is a $\Theta^{\pm\mathrm{ell}}$NF-Hodge theater in its own universe. The diagram does *not* commute [IUTchIII, Rem 1.4.1].

The single most important fact about this picture is that it does not commute. Going up then right is not the same as going right then up, because the log-link mixes the additive and multiplicative structures and the Θ-link only respects the multiplicative one. The Konami code does not work here: you may press ↑ as often as you like, but ↑→ and →↑ load different save files, and there is no ↓ at all (the log-link is not invertible). The toolkit's TUI lets you press `l` and `t` to make these moves and watch the universe counter climb, which is the closest thing we have to a sugoroku board for this theory.

**OTAKU NOTE: TIME LOOPS**

Time-loop anime teach that you can repeat the same week as often as you like, as long as you carry something across each reset. In the log-theta-lattice, every vertical step is a reset of the ring structure, and what you carry across is the étale-like data, the D-prime-strips. The protagonist does not remember the additive structure, but the fundamental group remembers for them. That is what "vertically coric" means.

## 5 Characteristic four: the theaters combine 合体！

Each dot in Figure 1 is assembled from two halves glued together [IUTchI, Def 6.13 (iii)]:

$$\mathcal{HT}^{\Theta^{\pm\mathrm{ell}}\mathrm{NF}} \;=\; \mathcal{HT}^{\Theta^{\pm\mathrm{ell}}} \bigsqcup_{0,\,\boxplus \mapsto >} \mathcal{HT}^{\Theta\mathrm{NF}}.$$

The left half, the $\Theta^{\pm\mathrm{ell}}$-Hodge theater, is *additive*: it carries a capsule of $l = 13$ prime-strips labelled by $t \in \mathbb{F}_l = \{-6, \ldots, 6\}$, acted on by $\mathbb{F}_l^{\times\pm}$, and two bridges, the $\Theta^{\pm}$-bridge to $\mathcal{F}_{\boxplus}$ and the $\Theta^{\mathrm{ell}}$-bridge to $\mathcal{D}^{\circledcirc\pm}$. The right half, the ΘNF-Hodge theater, is *multiplicative*: a capsule of $l^* = 6$ prime-strips labelled by $j \in \mathbb{F}_l^* = \{1, \ldots, 6\}$, acted on by $\mathbb{F}_l^*$, with the Θ-bridge to $\mathcal{F}_{>}$ and the NF-bridge to $\mathcal{F}_{\circledast}$. The gluing sends $\pm t \mapsto |t|$ and identifies the zero label and $\boxplus$ with $>$. The toolkit checks all four bridges and the gluing for the example and prints ✓ for each.

This is a sentai team-up episode. Thirteen rangers on the additive team, six on the multiplicative team, and a combination sequence in which every pair of mirror-image rangers $t$ and $-t$ fuses into one ranger $|t|$, while ranger zero becomes the cockpit. The resulting mecha is unique: the gluing has no non-trivial automorphisms compatible with the structure [IUTchI, Rem 6.12.2], so there is exactly one way to say the transformation phrase. Mochizuki's term for the additive symmetry, *conjugate synchronization*, is also a perfectly good name for a dance routine.

### The two pilots

Every Hodge theater carries two "pilot objects" [IUTchIII, Def 3.8]. The *q-pilot* is the collection $\{\underline{\underline{q}}_v\}$; the *Θ-pilot* is the collection of theta values $\{\underline{\underline{q}}_v^{j^2}\}_{1\le j\le l^*}$, that is $\underline{\underline{q}}^1, \underline{\underline{q}}^4, \underline{\underline{q}}^9, \underline{\underline{q}}^{16}, \underline{\underline{q}}^{25}, \underline{\underline{q}}^{36}$ for $l = 13$. Their log-volumes in the example are

$$\text{q-pilot: } -|\log(\underline{q})| = -0.9309, \qquad \Theta\text{-pilot (naive): } \tfrac{l(l+1)}{12}\cdot\big(-|\log(\underline{q})|\big) = -14.1180,$$

where the factor $l(l+1)/12 = 15.1667$ is the average of $j^2$ over $j = 1, \dots, l^*$. Getting into the robot is not optional in this genre, and the Θ-link is exactly the moment when the Θ-pilot on one side is told that it is now the q-pilot on the other. If the two log-volumes really had to agree, we would get $-15.17\,|\log q| = -|\log q|$, which is false, and that tension, made precise, is the inequality.

THE TOOLKIT SAYS

```
Θ-Hodge theater: Θ_v evaluated at j ∈ F_l^*:  q_^1, q_^4, q_^9, q_^16, q_^25, q_^36
q-pilot  {q_v}            log-vol -|log(q)| = -0.9309    (in F⊩_Δ)
Θ-pilot  {q_v^{j²}}_{j≤l*} naive log-vol  = -14.1180    (in F⊩_gau / F⊩_LGP)
```

It is a little like a monster-collecting game's evolution chart, where $\underline{\underline{q}}$ evolves into $\underline{\underline{q}}^4$, then $\underline{\underline{q}}^9$, and so on up to $\underline{\underline{q}}^{36}$, except that each evolution makes the monster *smaller*, because $|q_v| < 1$.

## 6 Characteristic five: one truth, from the fundamental group 真実はいつもひとつ

IUT is built on *anabelian geometry* (遠アーベル幾何), Grothendieck's idea that certain "hyperbolic" varieties are determined by their étale fundamental groups. Mochizuki proved a $p$-adic version of Grothendieck's conjecture for hyperbolic curves in the 1990s, and IUT relies on a stronger, *mono-anabelian* form: an explicit algorithm that rebuilds the field and the curve from the group alone. Every prime-strip in the theory comes in two flavours [IUTchI, Def 4.1, 5.2]:

- **étale-like** data, the *D-prime-strips*, which remember only the groups $\Pi_v$ (and $G_v$); these are what the log-link cannot change, so they are shared vertically;
- **Frobenius-like** data, the *F-prime-strips*, which also carry monoids such as $\mathcal{O}^\times$ and $\mathcal{O}^\triangleright$ that the log-link rearranges.

The two are connected by *Kummer theory*: a Frobenius-like monoid element is sent to its Kummer class in the cohomology of the étale-like group, which is how the theory reads Frobenius-like data from inside the étale-like world.

The genre here is the detective show. A famous boy detective reconstructs the culprit from a cufflink; the mono-anabelian algorithm reconstructs the number field from $\Pi_v$. Both insist there is only one truth. The Kummer map is the transformation belt from a long-running tokusatsu series: put the Frobenius-like monoid in, press the button, and a cohomology class comes out. Henshin! The D-prime-strip is what is left when you take every monoid away. In one of our earlier infographics, a bathhouse witch took the monoids and told the F-prime-strip that from now on its name was D; we stand by that joke.

FROM THE PAPERS: CORICITY

The D-prime-strips are *vertically coric*, invariant under the log-links, and the $\mathcal{F}^{\vdash\times\mu}$-prime-strips (units modulo torsion, with only mono-analytic structure) are *horizontally coric*, shared by the two sides of a Θ-link [IUTchIII, Thm 1.5, Def 3.8]. The whole of [IUTchIII] can be read as an effort to describe the Θ-pilot using only coric data, so that the description makes sense on both sides of the Θ-link at once. That description is the *multiradial representation* of Theorem 3.11.

## 7 Characteristic six: the price is indeterminacy 戦闘力測定

Describing the Θ-pilot with coric data is possible, but not for free. [IUTchIII] Theorem 3.11 gives the multiradial representation up to three *indeterminacies*:

- **(Ind1)** automorphisms of the étale-like data, coming from the fact that only the groups, not a choice of basepoint, are shared;
- **(Ind2)** automorphisms of the units modulo torsion $\mathcal{O}^{\times\mu}$, which the Θ-link cannot see;
- **(Ind3)** the *upper semi-compatibility* of the log-Kummer correspondence: going up and down the log-links only lets you say that one region contains another, not that they are equal.

Taking the union of all possible outcomes (the "holomorphic hull") and comparing log-volumes gives [IUTchIII] Corollary 3.12, the inequality $-|\log(\underline{\underline{\Theta}})| \geq -|\log(\underline{\underline{q}})|$, and [IUTchIV] turns it into the height bound of §1. The toolkit follows [EssLgc, §3.10], which breaks the argument into eight steps (Stp1)–(Stp8). For the example, the log-volume of the region being tracked is 33.0285 through the indeterminacy steps, 47.1465 after taking the hull, and the final comparison happens at $-|\log q| = -0.9309$.

Ind3 is the scouter in a famous martial-arts anime. It does not tell you a power level, it tells you a power level *is at least* something, and then it explodes. For our example, the toolkit's version of [IUTchIV] Theorem 1.10 produces the constant

$$C_\Theta \approx 1.88 \times 10^{22},$$

so the inequality holds with a very large amount of room. We have checked, and this number is over 9000. That is not evidence for or against anything: for $l = 13$ and a single triple the bound is trivially satisfied, because the theory is asymptotic. It speaks about bounded discrepancy classes, i.e. about infinitely many points at once, and the auxiliary prime $l$ is meant to grow with the height. One triple is a demo, not a test, the same way one episode is not a season.

## 8 Observations 感想戦

### 8.1 The author says the logic is simple

[EssLgc] (2023) summarizes the logical structure of IUT with Boolean operators. The Θ-link corresponds to logical AND $\wedge$; the indeterminacies coming from the log-Kummer correspondence correspond to exclusive OR $\dot{\vee}$:

$$A \wedge B \;=\; A \wedge (B_1 \,\dot{\vee}\, B_2 \,\dot{\vee}\, \cdots) \implies \cdots$$

and the paper compares the pattern to carry-addition in the Witt ring $\mathbb{Z}/4\mathbb{Z}$ written with Boolean $\wedge$ and $\dot{\vee}$ over $\mathbb{F}_2$. The April 2026 progress report on formalization [Form26] goes further and says that the logical structure "is rather simple", so that the value of a Lean formalization is mainly as a communication tool, "a precise record of the logical structure of IUT that is immune to false misinterpretations". The toolkit's numerical model of [EssLgc, Ex 2.4.5] reproduces the point in miniature: the AND version concludes "A = ε $\leq$ 1", while the OR, XOR and redundant-copies versions conclude "no conclusion".

We would describe this as a visual novel with one true ending. Choose $\wedge$ at every branch and you reach the inequality. Choose $\vee$, or identify two copies that the story wanted kept apart, and you get the bad ending where nothing follows. The disagreement of 2018 is, in this language, a disagreement about which route the original script is on.

### 8.2 The disagreement is about one corollary

It is worth being precise. Scholze and Stix did not claim that the papers are full of errors; they argued that one may identify the copies of the real numbers (and of the other objects) on the two sides of the Θ-link, since they are isomorphic, and that once one does so the argument for Corollary 3.12 only goes through after inserting scaling factors that make the resulting inequality useless [SS18]. Mochizuki's reply, worked out at length in [EssLgc, §3], is that such an identification is exactly what the theory forbids: the copies are "RCS-redundant" only in a reading that collapses the $\wedge$ of the Θ-link into a $\vee$. Outside RIMS and its collaborators the proof is not, as of this writing, generally accepted, and the formalization effort [Form26] is in part an attempt to settle the question in a form neither side can misread. We note this as a characteristic of IUT, not a verdict on it.

### 8.3 The notation is a light novel title

To give every object a name in plain ASCII, the toolkit needs 448 identifiers (`docs/IDENTIFIERS.md`). Some are short. Others, such as $\mathcal{F}^{\mid\!\blacktriangleright\times\mu}$-prime-strip → `FMonoGlobValUnitsModTorsPrimeStrip` or $\Theta^{\times\mu}{}_{\mathrm{LGP}}$-link → `ThetaUnitsModTorsLGPTorsorLink`, are as long as the title of a light novel, and say about as much about the plot. A fair English rendering of the latter would be "That Time I Got Linked Across Universes and Only My Units Modulo Torsion Made It". We believe this would sell.

### 8.4 There are effective versions, with very large constants

[ExpEst] (Mochizuki, Fesenko, Hoshi, Minamide, Porowski) makes the inequalities numerically effective by reworking the étale theta function so that it behaves at places above 2. One of its consequences (Theorem B) is an explicit Szpiro inequality $|abc| \leq 2^4 \cdot \max\{\exp(1.7 \cdot 10^{30} \cdot \varepsilon^{-166/81}),\ \mathrm{rad}(abc)^{3(1+\varepsilon)}\}$, and another (Corollary C) is the claim that Fermat's equation has no solutions for prime exponents $p > 1.615 \cdot 10^{14}$, which, combined with results of Vandiver, Coppersmith and Mihăilescu–Rassias, the authors present as an alternative proof of Fermat's Last Theorem. The constant $\exp(1.7 \cdot 10^{30})$ deserves its own isekai: it is the size of a world, not of a number.

### 8.5 What this repository can and cannot do

The toolkit in `src/` (C, with a TUI) and `htdocs/` (the same engine in WebAssembly) turns the objects of [IUTchI–IV] into data types, with generators, validators and 73 named processes, and `lean/` proves that the constructed objects satisfy the validators. It lets you build the example, walk the lattice, and watch the pilots and indeterminacies move. It does *not* verify Corollary 3.12: the anabelian reconstructions are modelled by

labelled placeholders, not computed, and the log-volumes are the numbers the papers predict, evaluated on the example. It is a tour of the set, not the film.

## 9 Character profiles キャラクター紹介

| Object | What it is | Its energy |
|---|---|---|
| Initial Θ-data | $(\overline{F}/F, \mathcal{X}_F, l, \underline{\mathcal{C}}_K, \underline{\mathbb{V}}, \mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}, \underline{\epsilon})$, the starting data [IUTchI, Def 3.1] | Character creation screen |
| Prime-strip | One local object for each place $v$, bundled together [IUTchI, Def 4.1, 5.2] | A bento: one compartment per prime |
| D-prime-strip | Étale-like: only the groups $\Pi_v$ | The ghost without the shell |
| F-prime-strip | Frobenius-like: groups plus monoids | The ghost with the shell, for now |
| Hodge theater | $\Theta^{\pm\mathrm{ell}}$-HT ⊔ ΘNF-HT, one per universe [IUTchI, Def 6.13] | Sentai combination, 13 + 6 rangers |
| Θ-link | Identifies Θ-pilot with q-pilot, multiplicatively only [IUTchIII, Def 3.8] | Truck-kun |
| log-link | $p$-adic logarithm to a log-shell [IUTchIII, Def 1.1] | A time-loop reset |
| Log-shell $\mathcal{I}_v$ | $(p_v^*)^{-1}\log_v(\mathcal{O}^\times_{K_v})$, the common container | The save file everyone can read |
| Kummer map | Frobenius-like monoid $\to$ étale-like cohomology | Transformation belt |
| Multiradial rep. | The Θ-pilot described by coric data [IUTchIII, Thm 3.11] | The recap episode everyone can follow |
| Ind1, Ind2, Ind3 | The three indeterminacies of Thm 3.11 | A scouter that only says “at least” |
| Cor 3.12 | $-\lvert\log \underline{\underline{\Theta}}\rvert \geq -\lvert\log \underline{\underline{q}}\rvert$ | The final boss, still contested |

**次回予告 · NEXT EPISODE**

The log-link climbs again! The Θ-pilot crosses into a new universe with nothing but its monoid and its degree. Can the multiradial representation describe it from the other side? Will Ind3 let it keep more than an upper bound? And will anyone, in any universe, finally formalize Corollary 3.12 in Lean? Next time on funfun IUT: “Upper Semi-Compatibility Is Not Equality”. Don't miss it!

### おまけ · Omake

Why did the $\Theta^{\pm\mathrm{ell}}$-Hodge theater refuse to go to the karaoke box? Because every song it chose had a mirror-image duet partner, and only $|t|$ was allowed on stage. Why is the log-theta-lattice a bad place to buy a train pass? Because every line goes up and to the right, and the transfers do not commute. Why does the Frey curve for $1 + 2400 = 2401$ never lose at mahjong? It only ever needs four sevens.

## ★ References 参考文献


- [IUTchI–IV] S. Mochizuki, *Inter-universal Teichmüller Theory I: Construction of Hodge Theaters; II: Hodge–Arakelov-theoretic Evaluation; III: Canonical Splittings of the Log-theta-lattice; IV: Log-volume Computations and Set-theoretic Foundations*, Publ. RIMS 57 (2021). In this repository as `pdf/Inter-universal Teichmuller Theory I-IV.pdf`.

- [Alien] S. Mochizuki, *The Mathematics of Mutually Alien Copies: from Gaussian Integrals to Inter-universal Teichmüller Theory*. `pdf/Alien Copies, Gaussians, and Inter-universal Teichmuller Theory.pdf`.
- [EssLgc] S. Mochizuki, *On the Essential Logical Structure of Inter-universal Teichmüller Theory in Terms of Logical AND "$\wedge$" / Logical OR "$\vee$" Relations* (September 2023). `pdf/Essential Logical Structure of Inter-universal Teichmuller Theory.pdf`.
- [ExpEst] S. Mochizuki, I. Fesenko, Y. Hoshi, A. Minamide, W. Porowski, *Explicit Estimates in Inter-universal Teichmüller Theory*. `pdf/Explicit estimates in IUTeich.pdf`.
- [Form26] S. Mochizuki, *On the Formalization of IUT: a Preliminary Progress Report* (joint work in progress with Y. Hoshi, G. Yamashita, Y. Yang, …), April 2026. `pdf/Formalization of IUT (2026-04).pdf`.
- [SS18] P. Scholze, J. Stix, *Why abc is still a conjecture*, manuscript (2018), discussed in [EssLgc, §1].
- [Toolkit] This repository: `src/`, `htdocs/`, `lean/`, `docs/DATA_MODEL.md`, `docs/PROCESSES.md`. Every number above was produced by `node scripts/funfun/build.js` from `src/iut` on $(1, 2400, 2401)$, $l = 13$.


Every show, game and character referred to above belongs to its creators and is mentioned only by description, with affection. ありがとうございました！